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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	COVID-19 / Public Health Data Visualization Dashboard	
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Abstract

The **COVID-19 / Public Health Data Visualization Dashboard** is a data science-based analytical system designed to analyze and visualize COVID-19 and public health data to identify important trends, patterns, and relationships. The project uses a comprehensive COVID-19 dataset containing attributes such as confirmed cases, active cases, recovered cases, deaths, vaccination rates, testing data, and geographical information. The system applies data preprocessing and exploratory data analysis to handle missing values, remove inconsistencies, understand data distributions, and identify relationships between different public health indicators. The analysis examines the spread and progression of COVID-19 across different regions and time periods, providing insights into infection rates, mortality, recovery, and vaccination trends.

For data analysis and visualization, the system applies statistical analysis and data visualization techniques to transform raw public health data into meaningful information. Various charts, graphs, time-series plots, and geographical visualizations are used to compare COVID-19 cases, deaths, recoveries, and vaccination rates across regions. The dashboard allows users to interact with the data by applying filters based on location and time period, making it easier to explore specific trends and compare public health conditions. The visualizations provide a clear representation of the progression of the pandemic and help identify regions and periods with significant changes in COVID-19 activity.

The selected analytical results are integrated into an interactive dashboard where users can explore key public health indicators and obtain meaningful insights from the dataset. The dashboard also presents statistical summaries, trend analysis, regional comparisons, and visual analytics to improve data interpretation. The primary objective is to demonstrate how **data science, exploratory data analysis, and data visualization** can be used to understand

large-scale public health datasets and support data-driven decision-making. The system is intended as an analytical and visualization tool for understanding COVID-19 trends and public health patterns and does not replace official medical, epidemiological, or public health guidance.

Signature of the Guide HOD